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A Novel Method of Audio Steganography using Advanced Encryption Standard

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Abstract: This paper introduces a new sound steganographic algorithm based on Advanced Encryption Standard (AES). First of all sound is converted in the form of an image and then it is encrypted with the help of AES. Simulations and security analyses are provided to demonstrate an excellent performance of the proposed algorithm.

Keywords: Sound; AES; image; encryption

1 Introduction

With the rapid advancement in network technologies and smart phone systems, a growing number of sounds, images and videos with private and confidential information are generated, transmitted and stored every minute. The increasing use of sounds, images and videos leads to the problem of how to prevent them from different types of security attacks and information leakage and to ensure the integrity of digital data. Therefore, the security of digital data has drawn researcher's attentions.

Many image encryption algorithms have been developed to protect images and videos in different fields [1–3]. Examples include traditional data encryption standards such as data encryption standard (DES) [4] and advanced encryption standard (AES) [5], and image encryption methods based on different technologies such as chaotic systems [6–8], random grids [9], wave transmission [10] and optical encryption techniques [11,12].

Among existing image encryption technologies, an interesting technique based on image bit plane decomposition has shown excellent encryption performance and is easily implemented in hardware [13]. This technique first decomposes an image into several binary bit planes using a specific decomposition method, such as traditional binary decomposition, gray code decomposition [14] and Fibonacci p-code decomposition [15–17]. It then manipulates bit planes using various methods, and combines bit planes to obtain the encrypted images. Based on this technique, a list of image encryption algorithms has been proposed recently. Han et al. encrypted bit planes of optical images using the exclusive-OR (XOR) operations [18]. Later, selective bit plane encryption schemes were developed for image encryption to achieve low computational complexity [19, 20]. However, these algorithms have a low level of security due to their predictable decomposition results and/or limited key spaces [16]. Our previous work proposed image encryption algorithms using the parametric decomposition methods including the (nkp)-Gray code decomposition [14] and Fibonacci p-code decomposition [16]. However, there still exists a room for security enhancement.

Steganography is a mechanism for hiding and retrieving the high sensitive information in data transmission. Steganography must not be confused with cryptography, where the message is converted into meaningless to malicious people who intercept it. The main goal of steganography techniques is that it is difficult to detect the image and so saved from attacks. The steganography approaches can be distinguished into three types: pure steganography, secret key steganography and public key steganography.

2 AES algorithm for Cryptography

Advanced Encryption Standard (AES) [5] is a standard for the encryption of electronic data. The U.S. government first time used it in 1997 and now use in worldwide. AES is a symmetric-key algorithm which means that the same key is used both of sender and receiver. This AES standard specifies the Rijndael algorithm [1], a symmetric block cipher that can process data blocks of 128 bits, using key size of 128, 192, and 256 bits. The input, the output and the cipher key are used in Rijndael. It takes an input and output of certain block size, only 128 bits.

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Advantages of using AES algorithm

1. Very Secure
2. Reasonable Cost
3. Main Characteristics:
   I. Flexibility, II. Simplicity

Proposed method

The proposed method to develop a sound steganography is shown in Figure 1. Basically the purpose of this method is to show the secure communication of sound signals via insecure channel with the help of image encryption applications. The weak point of this algorithm is that there is a minor difference between original sound and extracted sound. The steps of proposed scheme are as follows:

Step 1: Read sound file in Matlab with '*wavread*' command.

Step 2: Convert units of plaintext sound in the form of $GF(2^8)$ elements with $'uint8'$ command.

Step 3: Usually the sound has the form of column matrix, convert column matrix in the form of $M \times N$ matrix.

Step 4: Transform $M \times N$ matrix in the form of gray level image.

Step 5: Apply advance encryption standard to encrypt $M \times N$ image, because we need to secure sound when we will send it from insecure line of communication. With this step we will get an encrypted image.

Step 6: In step 6 we send encrypted image from channel and it will received on the receiving end.

Step 7: Decrypt $M \times N$ image with the help of advanced encryption standard.

Step 8 : Extract sound file from decrypted image.

In Table 1 we have shown an example to explain the first 4 steps of the proposed algorithm, the remaining steps depend on AES.

Analyses

The major problem with sound steganography is that when we extract our sound back from the image after decrypting we do not get the original sound exactly. In proposed algorithm we are not claiming that we get the ideal condition in extracting of original image. But we with the help of some useful analyses we can show that the difference between original and extracted sound is very small. To analyze the difference we use some criterions such that mean square error (MSE), graphical spectroscopy analyses and peak signal to noise ratio.

Table 1. The matlab code corresponding to first 4 steps is as follows.

```
clc
clear
b = wavread('Hum.wav');
C = b;
D = C(1:50000);
x = 1;
y = 200;
ll = wavread('Hum.wav');
plot(ll,'g');
grid on
title ('wave form of first 50000 samples');
% code to convert the column matrix in M * N matrix
for i = 1:250
   L = D(x:y);
   N = L';
 A(i,:) = N;
 x = x + 200;
 y = y + 200;
end
imwrite(A,'projectpic.jpg');
wavwrite(D,42100,'project.wav');
projectpic = imread('projectpic.jpg');
% Now convert to double type
projectpic = im2double(projectpic);
jpg = [;];
for i = 1:250
   w = projectpic(i,:); m = w';
   jpg = vertcat(jpg,m);
end
```

Mean square error

Mean Squared Error (MSE) is one of the earliest tests that were performed to test if two sounds are similar. A function could be simply written according to equation (1) and matlab code is also given below

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(X_i - X_i^*)^2 \tag{1}$$

Matlab code for MSE

```
function out = MSE(sound1,sound2)
e = 0;
[m,n] = size(sound1);
for i = 1 : m
for j = 1 : n
e = e + double((sound1(i,j) - sound2(i,j)) ^2);
end
end
out = e/(m * n);
end
```

PSNR

The Pick Signal to Noise Ratio (PSNR) is a better test since it takes the signal strength into consideration (not only the error). Below equation (2) and matlab code describes how this value is obtained:

$$PSNR = 10.\log_{10}\left(\frac{MAX_i^2}{MSE}\right) \tag{2}$$

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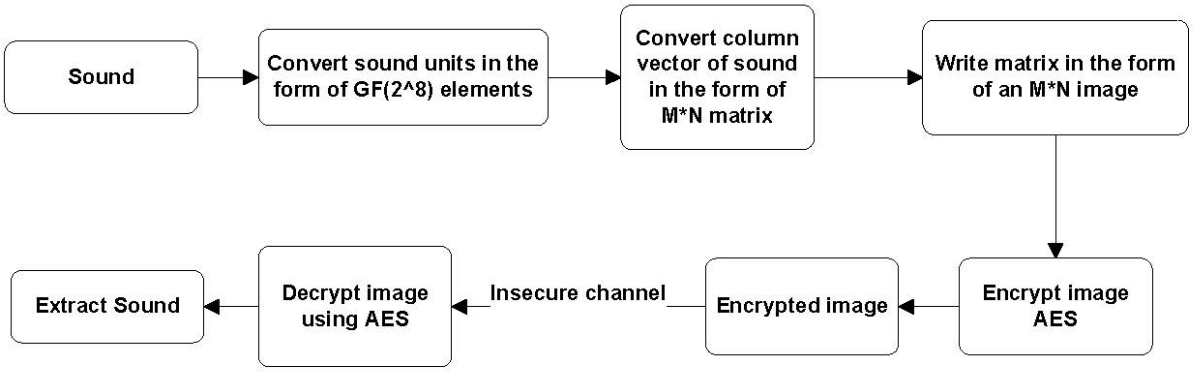


Fig. 1. Flowchart of propose algorithm.

Matlab code for PSNR

```
function out = PSNR(sound1,sound2)
e = MSE(sound1,sound2);
m = max(max(sound1));
out = 10 * log(double(m)^2/e);
end
```

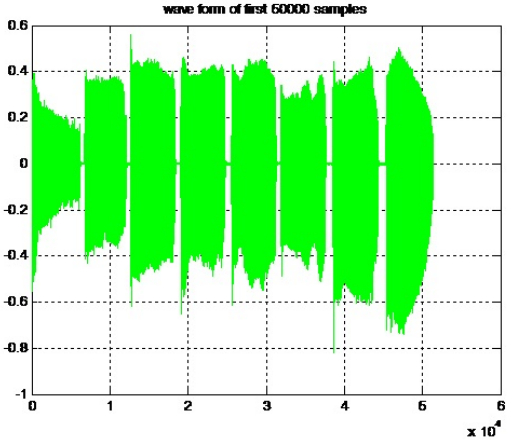


Fig. 2. Graphical representation of plaintext sound.

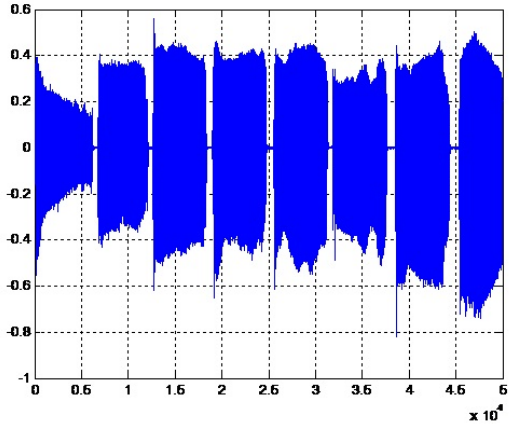


Fig. 3. Graphical representation of extracted sound.

Table 2. The values of MSE and PSNR show that the proposed technique is a good technique for steganography.

Analyses	Difference between original and extracted sound
MSE	8.345
PSNR	48.253

3 Conclusion

In this paper, we have introduced a method of imperceptible audio data hiding. Audio data hiding techniques can be used for a number of purposes other than to convert communication or deniable data storage, information tracing and finger printing, tamper detection. The technique presented in this paper is not ideal because there is a difference between the original voice and extracted voice we can observe this fact from Table 2 and figure 2–7. But what we gain in this technique that the difference is not too much and this technique is very efficient to use in small applications, and steps are also very simple. Basically here in this paper we have developed a relation between audio processing and image processing so one can apply any encryption algorithm to improve the security of image and audio as well.

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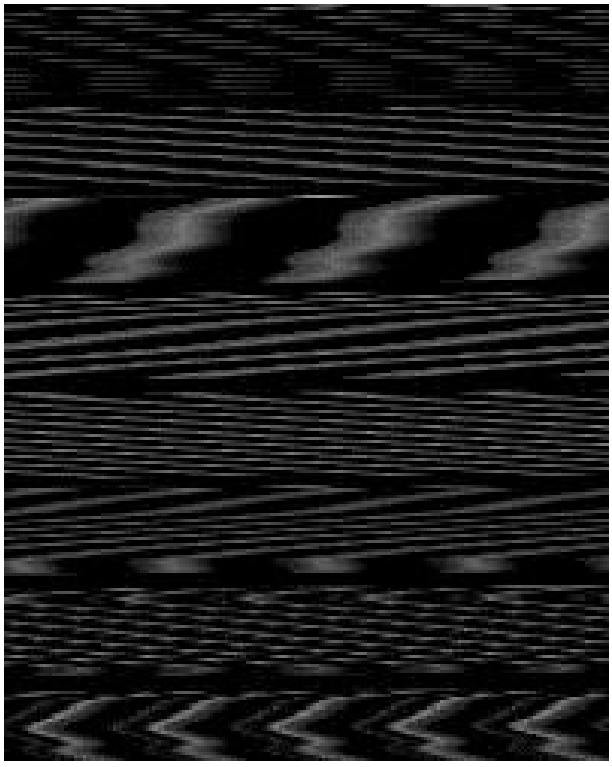


Fig. 4. Image corresponding to Figure 1 sound.

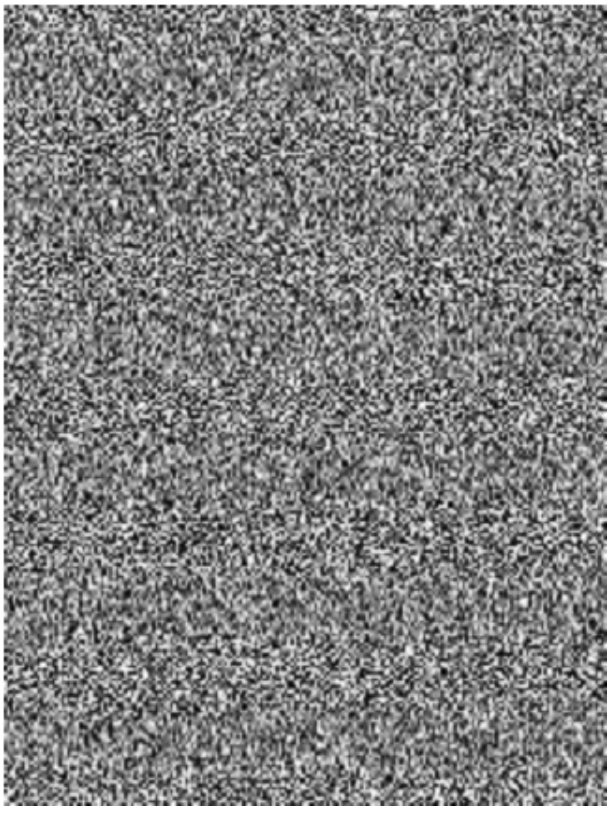


Fig. 5. Encryption of image using AES.

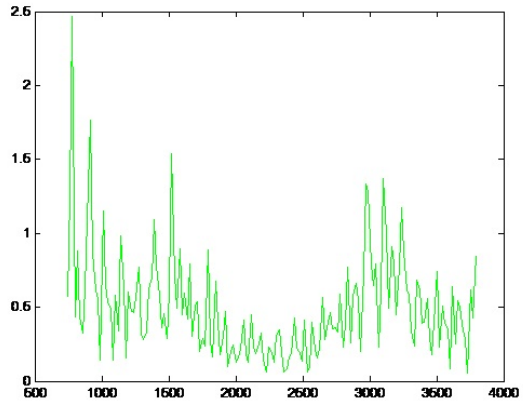


Fig. 6. Spectrometer readings corresponding to original sound.

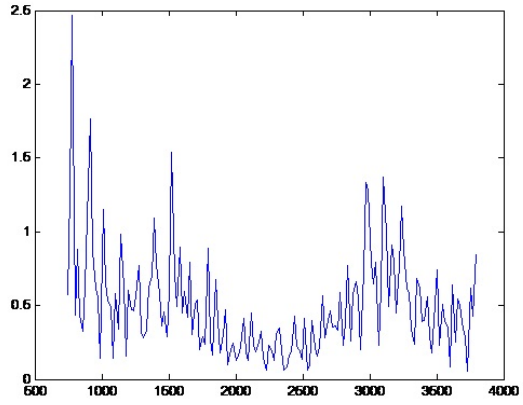


Fig. 7. Spectrometer readings corresponding to extracted sound.

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